Impact of Mutations in the CHK2 FHA Domain and N-Lobe of Kinase Domains on CHK2 Structure and Dynamics Through Molecular Dynamics Simulations

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3RD INTERNATIONAL GRADUATE RESEARCH SYMPOSIUM (IGRS'24)

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The Content of Proposal

1.INTRODUCTION
2.AIM
3.DESIGN AND METHODS
4.RESULTS
5.CONCLUSION

- To investigate the pathogenicity of variants (VUS) of unknown clinical significance
- To perform dynamic simulation of the effects of variants in 2 genes containing VUS on protein structure and dynamics

INTRODUCTION

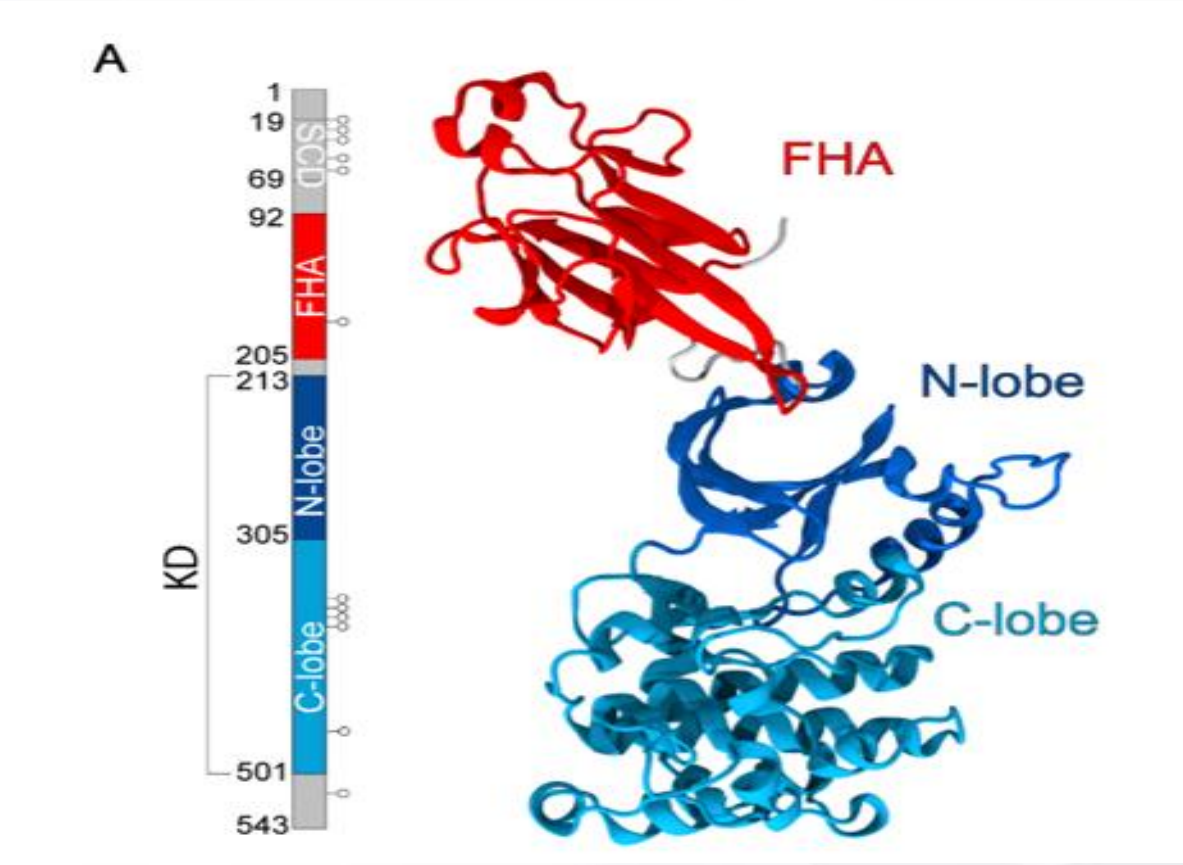


Figure 1. Structure of the monomer CHK2 protein

(PDB ID:3I6U, 2CN5; Cai et al., 2009, Oliver et al., 2006)

- The *CHEK2* gene encodes the serine/threonine kinase protein CHK2, which regulates DNA repair pathways and cell cycle checkpoints to maintain genomic integrity and stability.
- Upon DNA damage in cells, CHK2 facilitates repair of damaged DNA and inhibits cell division, thus preventing cancer development

AIM

- Effects of mutations on CHK2 protein structure and dynamics with Moleculer dynamic (MD) simulations

Simulation details

- ADP+P structure with wild type and mutant type simulations.
- All MD simulations were performed for NPT ensemble in explicit solvent using NAMD 3.0alpha9 (J.C. Philips et al., 2005) with CHARMM36 force field (Best et al., 2012)
- Simulations were performed 310 K temperature and 1 bar pressure.
- VMD was used for the sytem preparation.
- Mutations were perform using the VMD mutater plugin.

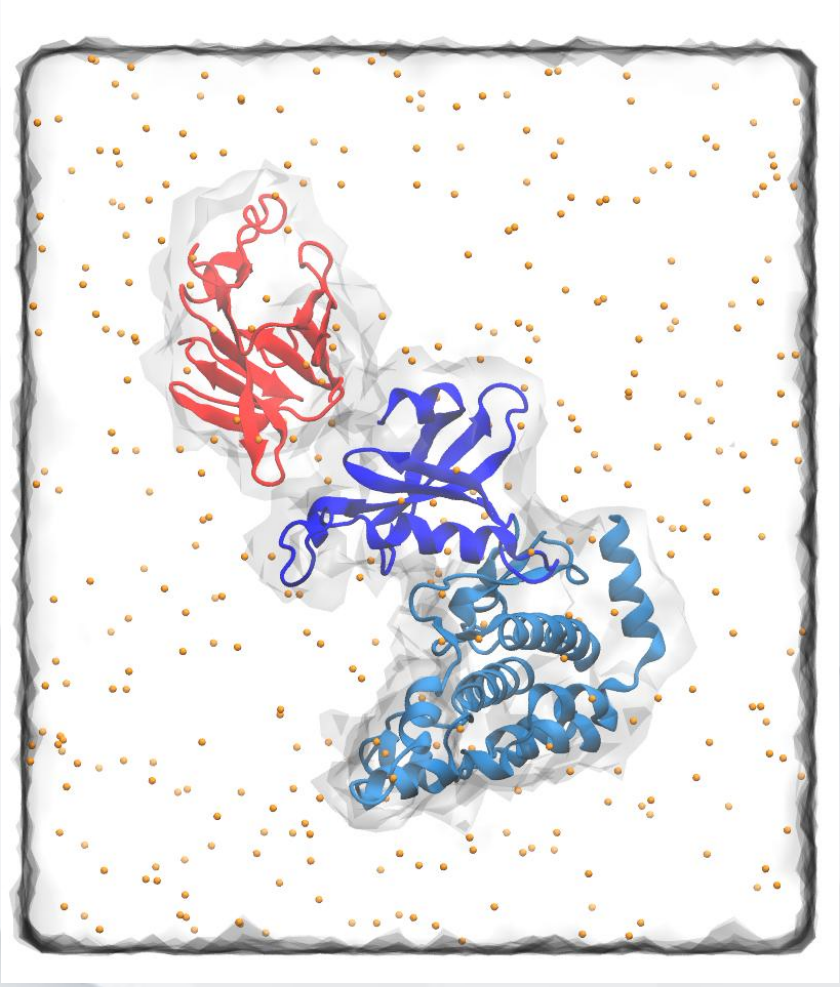


Figure 2. CHK2 protein

(PDB ID:3I6U, 2CN5; Cai et al., 2009, Oliver et al., 2006

RESULTS

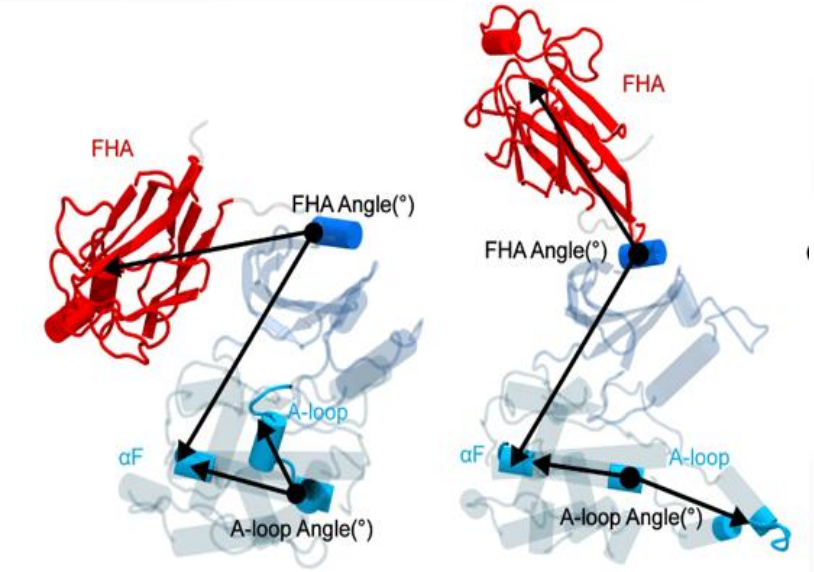
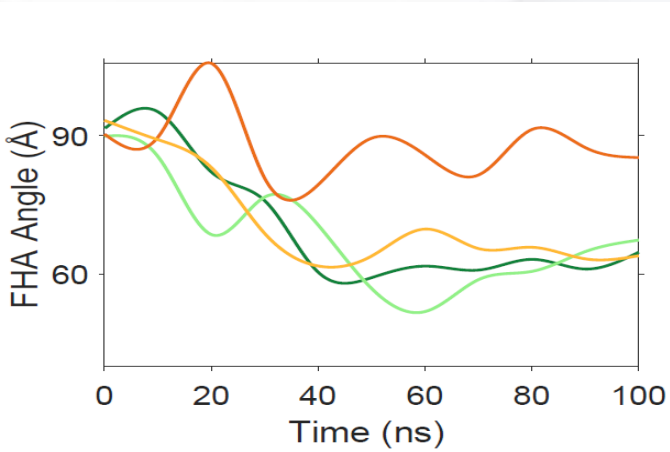
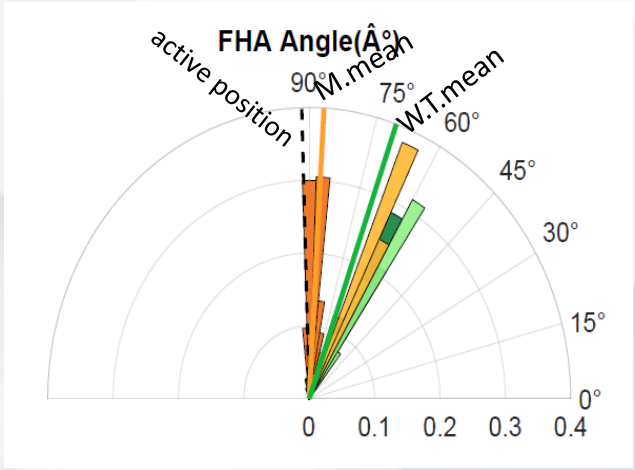


Figure 3. The angle defined for FHA and A-loop opening/closing are shown on the CHK2 structure

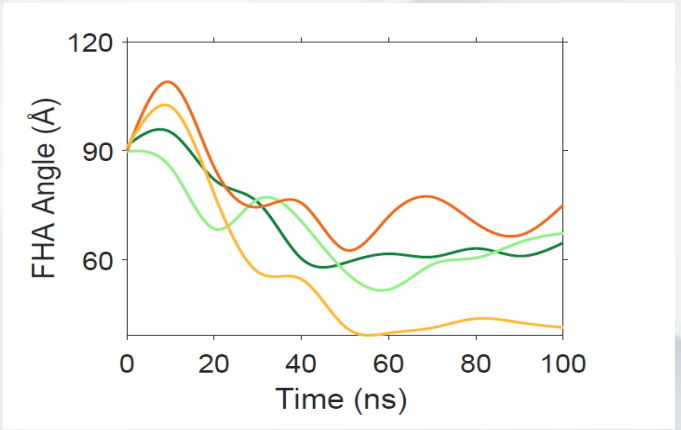
FHA Angle

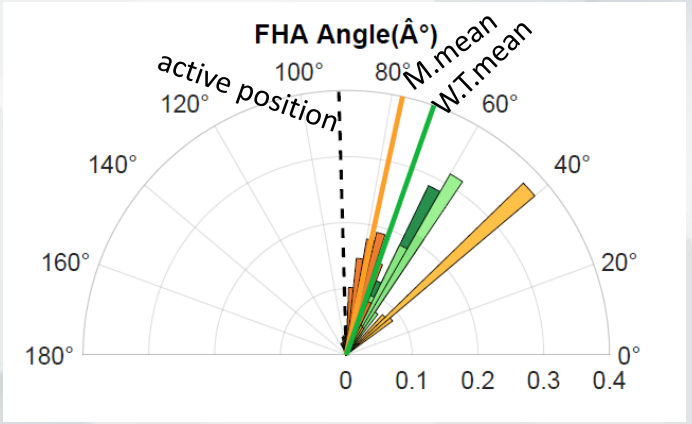
E320K





V246L





Wild type

Mutant

RESULTS

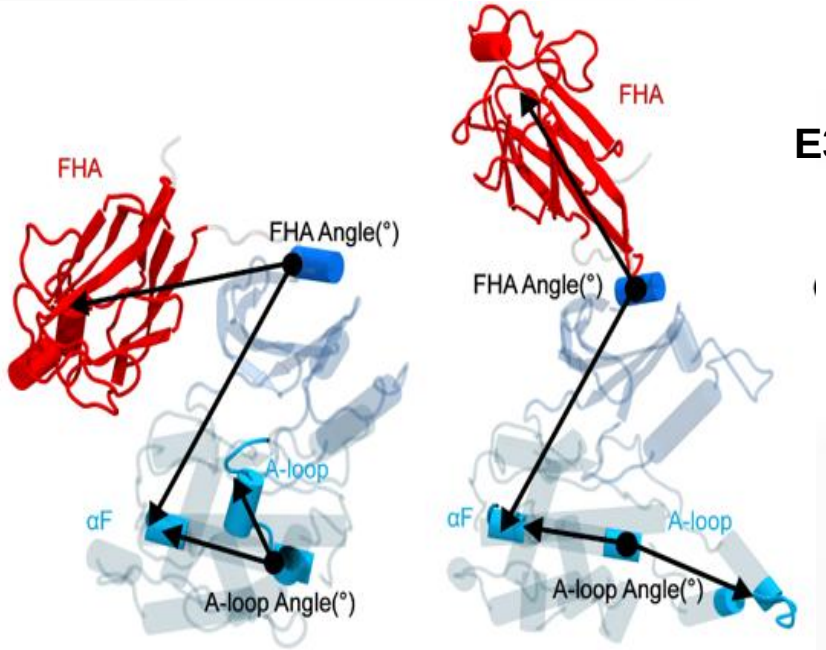
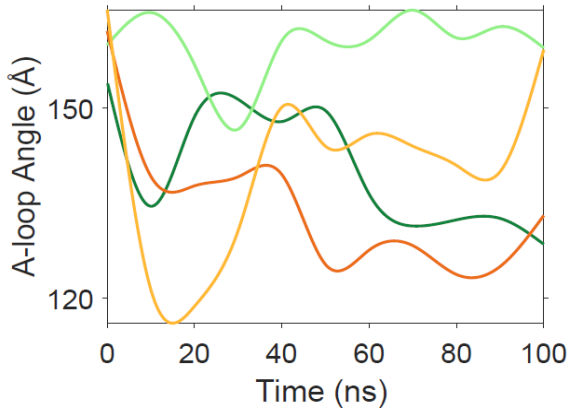
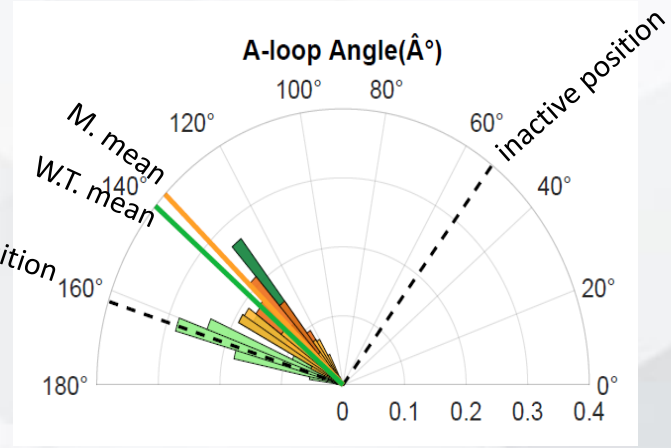


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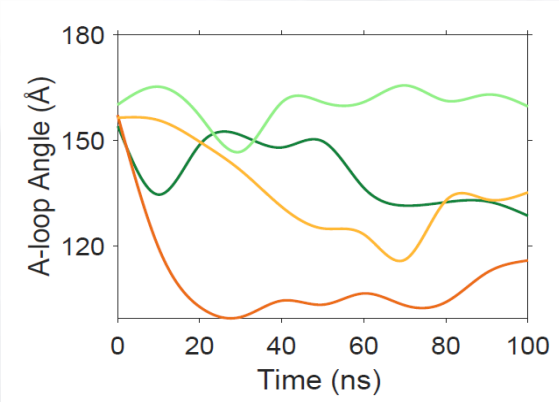
A-loop Angle

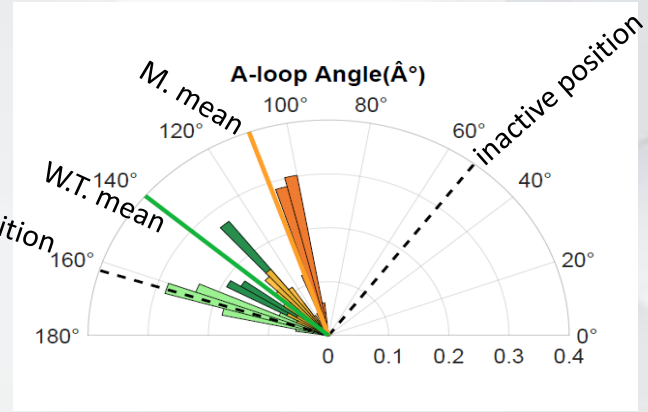
E320K





V246L



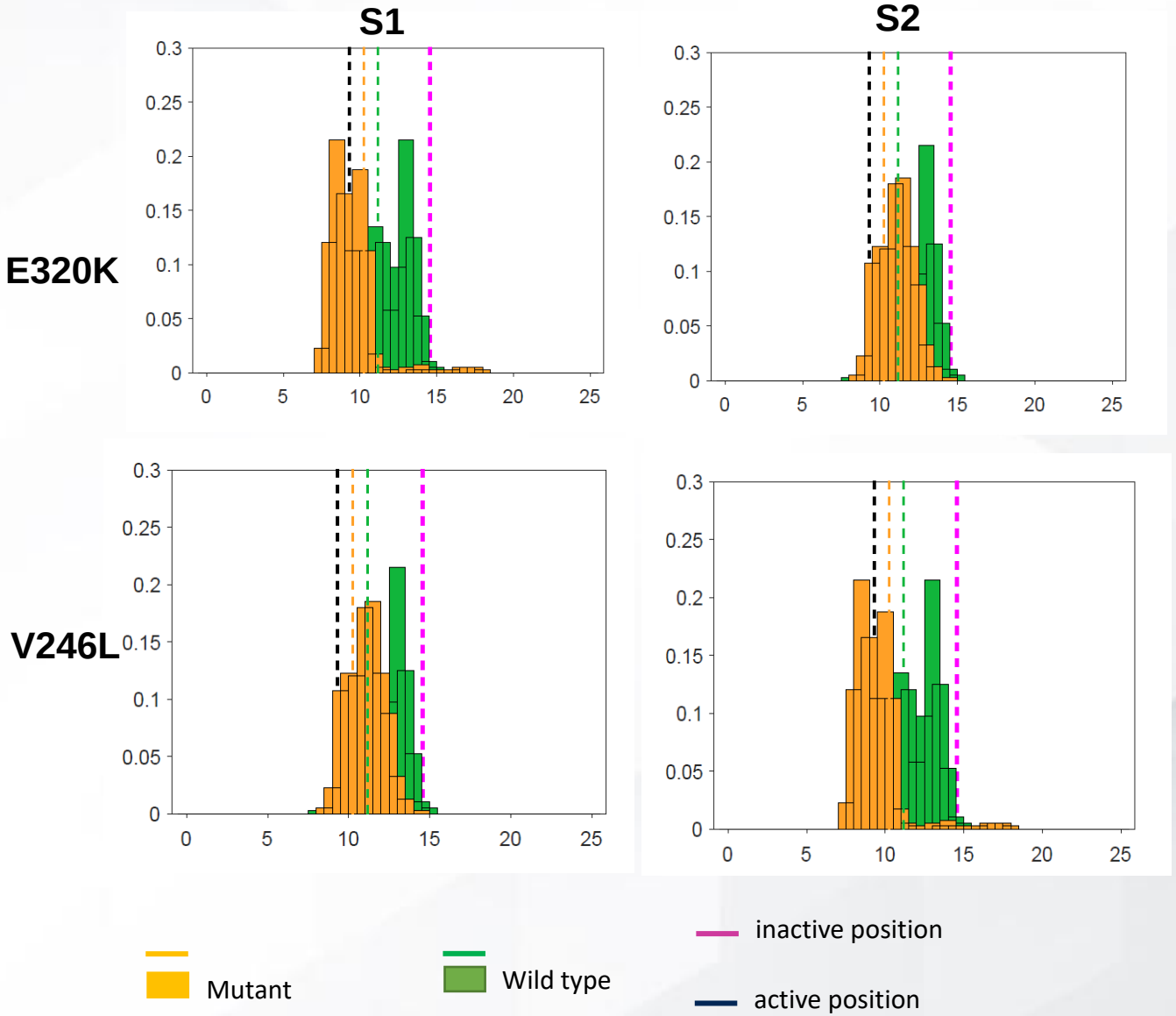


Wild type

Mutant

Figure 4. On the A-loop angle graphs, two dashed lines indicate different crystal structures: the higher degree line corresponds to 2CN5 with an extended A-loop, while the lower degree line corresponds to 3I6U with an A-loop in the "in" conformation.

RESULTS



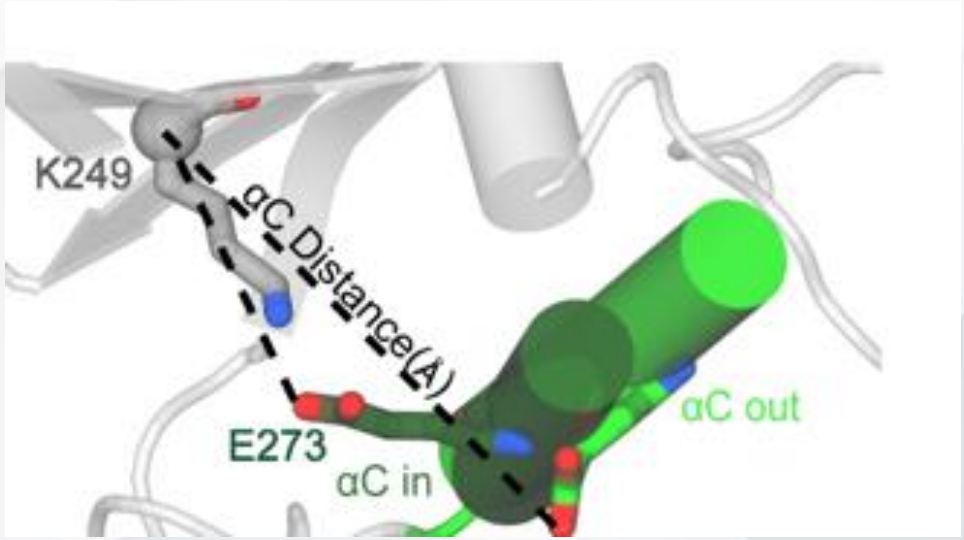


Figure 5.The distance defined for αC helix in/out conformations is shown

RESULTS

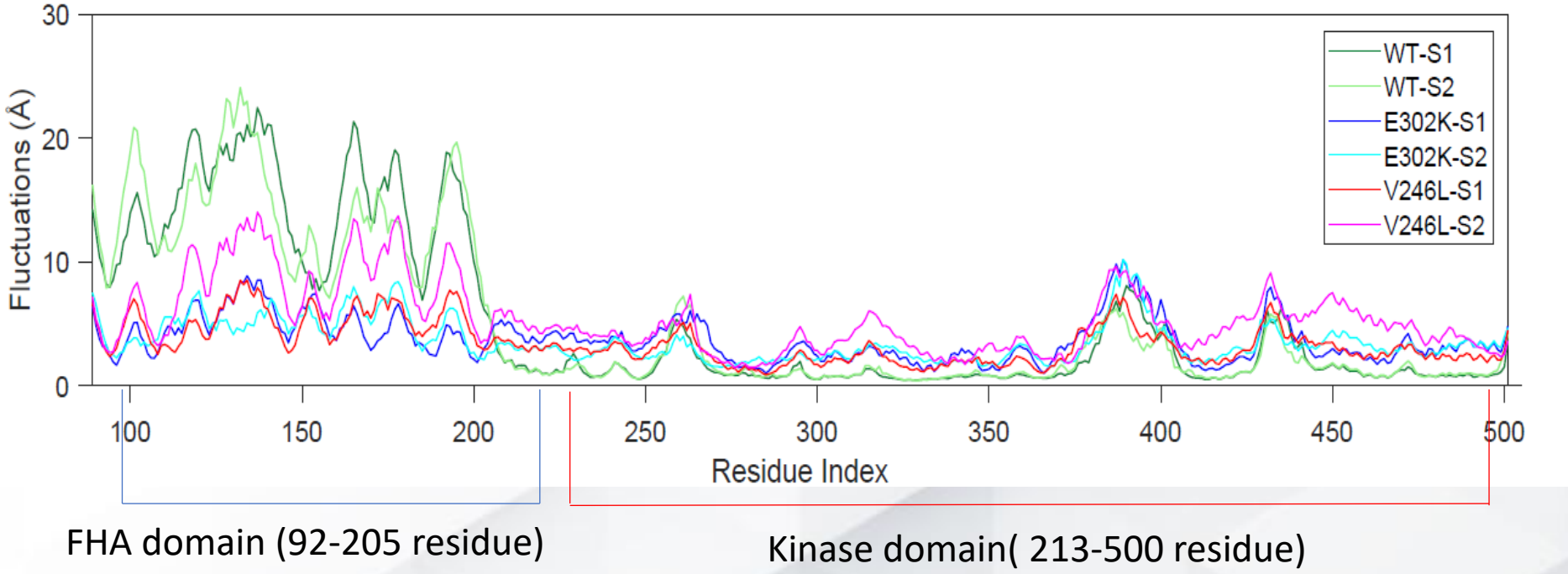


Figure 6. RMSF fluctuation of CHK2 protein after 100 ns simulation

CONCUSION

- Both mutations affected the structura and dynamics of CHK2
- The effect of the mutatıon on structure and dynamıcs was more pronopunced Mutant 246.
- Mutatıon 246 shıfted the proteın ınto a more actıve-lıke structure, represented with an incrased a loop angle and decreased FHA angle.
- Mutation 246 shıfted into a more actıve like-like structure
- Longer MD sımualtıons will provide better sampling and a more ındepth understanding of the effect of the mutations.

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